

AP CALCULUS AB - UNIT 1

1.12 Confirming Continuity over an Interval

Lesson Overview

- Distinguish continuity on open, closed, and half-open intervals.
- Use one-sided continuity at included endpoints.
- Find maximal intervals of continuity from domain restrictions.
- Justify continuity of sums, products, quotients, roots, and compositions on stated intervals.

Key Knowledge and Vocabulary

Interior points. Use ordinary two-sided continuity at every interior point.

Included left endpoint. Require $\lim_{x \rightarrow a^+} f(x) = f(a)$.

Included right endpoint. Require $\lim_{x \rightarrow b^-} f(x) = f(b)$.

Maximal interval. An interval cannot be enlarged without crossing a domain restriction or discontinuity.

Domain first. Continuity of a standard function is asserted only on its domain.

Explanation and Strategy

Step 1. Find the real domain before writing intervals.

Step 2. Split at denominator zeros, logarithm boundaries, even-root restrictions, and piecewise joins.

Step 3. Use square brackets only when the endpoint is in the domain and the correct one-sided continuity condition holds.

Solved Examples

Worked Example: Rational intervals

Find the maximal intervals of continuity of $f(x) = \frac{x+1}{x^2-4}$.

Solution. The denominator vanishes at ± 2 . The maximal intervals are $(-\infty, -2)$, $(-2, 2)$, and $(2, \infty)$.

Worked Example: Square-root endpoint

On what interval is $g(x) = \sqrt{5-x}$ continuous?

Solution. The domain is $x \leq 5$. Square root and the inner linear function are continuous, so g is continuous on $(-\infty, 5]$, with left continuity at 5.

Worked Example: Combined restrictions

Find maximal intervals for $h(x) = \frac{\ln(x+4)}{x^2-9}$.

Solution. Require $x > -4$ and $x \neq \pm 3$. Thus the intervals are $(-4, -3)$, $(-3, 3)$, and $(3, \infty)$.

Leveled Practice

Easy Level

Foundational fluency. Focus on notation, definitions, and direct procedures.

1. Find the intervals of continuity of $f(x) = 1/(x - 5)$.

2. Find the interval of continuity of $g(x) = \sqrt{x + 2}$.

3. Find the interval of continuity of $h(x) = \ln x$.

4. What one-sided condition is required at the right endpoint b of $[a, b]$?

Medium Level

Apply the idea in one or two connected steps.

5. Find maximal intervals for $\frac{x^2+1}{(x-1)(x+4)}$.

6. Find the continuity set of $\sqrt{9 - x^2}$.

7. Find maximal intervals for $\tan x$.

8. Is $\sqrt{x}$ continuous on $[0, 4]$? State the endpoint reasoning.

Hard Level

Connect representations, conditions, and justification.

9. Find maximal intervals for $f(x) = \frac{\sqrt{x+1}}{x-3}$.

10. Find maximal intervals for $g(x) = \sqrt{\frac{x-1}{x+2}}$.

11. Find maximal intervals for $h(x) = \frac{1}{\sqrt{x-2}}$.

12. Find maximal intervals for $p(x) = \frac{\sqrt{9-x^2}}{x-1}$.

Difficult Level

Use nonroutine structure and precise mathematical reasoning.

13. Find maximal intervals for $F(x) = \ln\left(\frac{x-2}{x+1}\right)$.

14. Determine where $G(x) = \arccos(2x - 1)$ is continuous.

15. Let f be continuous on $[-2, 3]$ and g continuous on $[1, 10]$. Give a sufficient interval on which $g(f(x))$ is continuous.

Challenge Level

Synthesize, construct, generalize, or prove.

16. Prove that $\max\{f, g\}$ is continuous wherever f and g are continuous.

17. Construct a function continuous on every interval $(-\infty, n)$ for integer n but not continuous on all of $\mathbb{R}$, or explain why impossible.

AP-Style Practice

Multiple-Choice Practice – No Calculator

Choose the best answer. Use exact values unless a decimal approximation is requested.

1. The maximal intervals of continuity of $1/(x^2 - 9)$ are

- (A) $(-\infty, \infty)$ (B) $(-\infty, -3) \cup (-3, 3) \cup (3, \infty)$ as three intervals
(C) $[-3, 3]$ (D) $(0, \infty)$

2. $\sqrt{4-x}$ is continuous on

- (A) $(-\infty, 4]$ (B) $[4, \infty)$
(C) $(-\infty, 4)$ only (D) $\mathbb{R}$

3. At the left endpoint a of $[a, b]$, continuity requires

- (A) a left-hand limit (B) a right-hand limit equal to $f(a)$
(C) both sides outside the domain (D) a derivative

Multiple-Choice Practice – Calculator Required

Choose the best answer. Use exact values unless a decimal approximation is requested.

4. A calculator domain plot for $\sqrt{(x-1)/(x+2)}$ suggests real values on

- (A) $(-2, 1)$ (B) $(-\infty, -2) \cup [1, \infty)$
(C) $[-2, 1]$ (D) $\mathbb{R}$

5. A graphing utility appears to connect across a denominator zero. What should determine the continuity intervals?

- (A) The drawn pixels only (B) The exact domain and algebraic restrictions
(C) The y-intercept (D) The color of the curve

Free-Response Practice – No Calculator

Let $f(x) = \frac{\sqrt{6-x}}{x^2-1}$.

(a) Find the real domain. [2 points]

(b) State the maximal intervals of continuity. [2 points]

(c) Explain the bracket at 6 and the parentheses at ± 1 . [2 points]

Free-Response Practice – Calculator Required

A calculator is used to inspect $g(x) = \ln(x+5)/(x-2)$.

(a) Use the graph/domain screen to conjecture the continuity intervals. [2 points]

(b) Confirm the intervals analytically.

[3 points]

(c) Why is -5 not included even though $\ln(x + 5)$ may decrease without bound there?

[1 points]